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MongoDB Practicals – Final Answers:

Lab 1 – Connection:

URI for local connection:

mongodb://localhost:27017

Lab 2 – Create Database & Collection

- Database: **finance_db**
- Collection: **transactions**

Lab 3 – Insert Document

```
{
```

```
  "txnId": "T1001",
```

```
"accountId": "A5001",
"type": "Credit",
"amount": 2500.75,
"currency": "INR",
"date": "2025-09-29T10:15:30Z",
"status": "SUCCESS",
"channel": "MobileBanking",
"remarks": "Salary credit",
"address": {
  "city": "Mumbai",
  "state": "Maharashtra",
  "country": "India"
},
"tags": ["monthly", "salary", "priority"]
}
```

Lab 4 – Import Data:

Lab 5 – Read / Query Documents (8 Queries Total)

① Find all Credit transactions

```
{ "type": "Credit" }
```

2 Find all transactions where amount > 2000

```
{ "amount": { "$gt": 2000 } }
```

3 Find transactions done in Mumbai (embedded document)

```
{ "address.city": "Mumbai" }
```

4 Find all transactions tagged as “salary”

```
{ "tags": "salary" }
```

5 Find transactions that have both “food” and “shopping” tags

```
{ "tags": { "$all": ["food", "shopping"] } }
```

6 Find transactions that contain any of these tags ("food" or "shopping")

```
{ "tags": { "$in": ["food", "shopping"] } }
```

7 Find transactions where array contains exactly 3 elements

```
{ "tags": { "$size": 3 } }
```

8 Find all Debit transactions

```
{ "type": "Debit" }
```

Lab 6 – Update Documents

```
db.transactions.updateOne(  
  { "remarks": "Travel Booking" },  
  {  
    $set: { "status": "FAILED", "remarks": "Hotel confirmation" },  
    $push: { "tags": "bonus" }  
  }  
)
```

Lab 7 – Delete Documents

```
db.transactions.deleteOne({ "txnId": "T1006" })
```

Lab 8 – Aggregation (Bonus)

Stage 1 (Match):

```
{ "type": "Debit" }
```

Stage 2 (Group):

```
{ "_id": "$accountId", "totalDebits": { "$sum": "$amount" } }
```

Summary these all Final 8 Queries :

1. { "type": "Credit" }
2. { "amount": { "\$gt": 2000 } }
3. { "address.city": "Mumbai" }
4. { "tags": "salary" }
5. { "tags": { "\$all": ["food", "shopping"] } }
6. { "tags": { "\$in": ["food", "shopping"] } }
7. { "tags": { "\$size": 3 } }
8. { "type": "Debit" }

